Results for XES3G5M-sub-small

Three training-graph variants; six models per variant; N = 5, 10, ..., 50.

Higher is better for Ada, NOV and Ep. Bold = best; underline = second best within each row.

Rankings use unrounded values. All 18 model runs completed for this dataset.

Training-graph variant	Retained exfr edges	Share of source	Mean forgetting
1. Remove exfr0.00 training edges	694,068 / 1,167,562	59.4%	0.506
2. Weighted nonzero sampling, up to 400/student	290,800 / 1,167,562	24.9%	0.644
3. Weighted nonzero sampling, up to 200/student	145,400 / 1,167,562	12.5%	0.657

Main model across the three variants

Variant	Ada@10	Ada@50	NOV@50	Ep0.5@50	Ep0.7@50
Remove exfr0.00 training edges	0.859	0.861	1.000	0.337	0.300
Weighted nonzero sampling, up to 400/student	0.862	0.860	1.000	0.422	0.376
Weighted nonzero sampling, up to 200/student	0.859	0.864	1.000	0.348	0.267

Group 1: Remove exfr0.00 training edges

Retained exfr edges: 694,068 / 1,167,562 (59.4%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.857	0.823	**0.900**	0.843	0.872	0.868
10	0.859	0.826	**0.877**	0.845	0.866	0.859
15	0.859	0.825	**0.877**	0.843	0.865	0.861
20	0.860	0.825	**0.873**	0.843	0.864	0.862
25	0.862	0.818	**0.875**	0.843	0.864	0.861
30	0.862	0.824	**0.875**	0.843	0.864	0.859
35	0.862	0.825	**0.876**	0.844	0.860	0.855
40	0.861	0.830	**0.873**	0.844	0.861	0.855
45	0.861	0.832	**0.874**	0.844	0.860	0.857
50	0.861	0.835	**0.874**	0.844	0.859	0.857

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**1.000**	**1.000**	1.000	0.999	**1.000**	**1.000**
10	**1.000**	**1.000**	0.999	0.999	**1.000**	**1.000**
15	1.000	**1.000**	0.998	0.999	1.000	**1.000**
20	1.000	**1.000**	0.998	0.999	1.000	**1.000**
25	1.000	**1.000**	0.998	0.999	1.000	**1.000**
30	1.000	**1.000**	0.999	0.999	1.000	**1.000**
35	1.000	**1.000**	0.999	0.999	1.000	**1.000**
40	1.000	1.000	0.999	0.999	1.000	**1.000**
45	1.000	1.000	0.998	0.999	1.000	**1.000**
50	1.000	1.000	0.991	0.999	1.000	**1.000**

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: Remove exfr0.00 training edges

Retained exfr edges: 694,068 / 1,167,562 (59.4%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.227	-0.040	-0.064	0.227	**0.370**	0.244
10	0.386	0.148	-0.057	0.302	**0.393**	0.374
15	0.437	-0.034	-0.096	0.348	**0.450**	0.409
20	0.361	0.365	-0.018	0.365	**0.459**	0.411
25	0.430	**0.455**	0.197	0.387	0.433	0.385
30	0.377	0.462	0.182	0.392	**0.464**	0.340
35	0.463	**0.467**	0.249	0.390	0.441	0.378
40	0.406	**0.510**	0.258	0.392	0.472	0.382
45	0.364	**0.531**	0.350	0.384	0.444	0.432
50	0.337	**0.437**	0.222	0.381	0.387	0.391

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.223	-0.043	-0.064	0.219	**0.367**	0.242
10	0.384	0.146	-0.057	0.294	**0.391**	0.373
15	0.435	-0.035	-0.097	0.341	**0.448**	0.407
20	0.359	0.364	-0.020	0.359	**0.458**	0.409
25	0.429	**0.454**	0.189	0.382	0.432	0.383
30	0.376	0.461	0.178	0.389	**0.463**	0.338
35	0.462	**0.466**	0.244	0.387	0.440	0.376
40	0.406	**0.509**	0.255	0.390	0.471	0.381
45	0.363	**0.531**	0.340	0.383	0.444	0.432
50	0.337	**0.437**	0.214	0.380	0.387	0.391

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: Remove exfr0.00 training edges

Retained exfr edges: 694,068 / 1,167,562 (59.4%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.208	-0.047	-0.064	0.205	**0.354**	0.234
10	0.373	0.139	-0.057	0.277	**0.378**	0.360
15	0.424	-0.042	-0.110	0.321	**0.437**	0.394
20	0.348	0.356	-0.055	0.341	**0.442**	0.396
25	0.418	**0.446**	0.131	0.364	0.418	0.369
30	0.365	**0.454**	0.115	0.370	0.449	0.321
35	0.452	**0.460**	0.171	0.369	0.426	0.359
40	0.397	**0.505**	0.187	0.373	0.456	0.365
45	0.356	**0.527**	0.259	0.365	0.431	0.417
50	0.330	**0.434**	0.152	0.365	0.375	0.375

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.186	-0.064	-0.064	0.173	**0.324**	0.219
10	**0.348**	0.117	-0.063	0.224	0.342	0.341
15	**0.396**	-0.065	-0.141	0.256	0.392	0.367
20	0.319	0.330	-0.112	0.267	**0.387**	0.364
25	0.389	**0.407**	0.065	0.288	0.366	0.334
30	0.335	**0.410**	0.039	0.292	0.398	0.286
35	**0.420**	0.409	0.092	0.293	0.373	0.322
40	0.364	**0.458**	0.098	0.295	0.402	0.328
45	0.326	**0.475**	0.175	0.287	0.379	0.382
50	0.300	**0.383**	0.072	0.283	0.323	0.342

Group 2: Weighted nonzero sampling, up to 400/student

Retained exfr edges: 290,800 / 1,167,562 (24.9%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.857	0.829	**0.900**	0.847	0.859	0.859
10	0.862	0.820	**0.877**	0.847	0.868	0.860
15	0.863	0.825	**0.877**	0.846	0.868	0.860
20	0.864	0.836	**0.877**	0.845	0.869	0.859
25	0.864	0.825	**0.875**	0.846	0.869	0.858
30	0.864	0.822	**0.875**	0.845	0.867	0.857
35	0.864	0.824	**0.875**	0.846	0.866	0.855
40	0.864	0.831	**0.874**	0.845	0.865	0.854
45	0.863	0.836	**0.874**	0.846	0.864	0.852
50	0.860	0.832	**0.874**	0.846	0.862	0.851

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**1.000**	**1.000**	1.000	1.000	**1.000**	**1.000**
10	**1.000**	**1.000**	0.999	1.000	**1.000**	**1.000**
15	**1.000**	**1.000**	0.998	0.999	**1.000**	**1.000**
20	**1.000**	1.000	0.998	0.999	**1.000**	**1.000**
25	1.000	1.000	0.998	0.999	**1.000**	**1.000**
30	1.000	1.000	0.999	0.999	**1.000**	1.000
35	1.000	1.000	0.999	0.999	**1.000**	1.000
40	1.000	1.000	0.999	0.999	**1.000**	1.000
45	1.000	1.000	0.996	0.999	**1.000**	1.000
50	1.000	1.000	0.991	0.999	**1.000**	1.000

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: Weighted nonzero sampling, up to 400/student

Retained exfr edges: 290,800 / 1,167,562 (24.9%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.231	0.000	-0.064	0.194	0.207	**0.240**
10	0.364	-0.125	-0.068	0.257	0.307	**0.383**
15	**0.456**	0.127	-0.009	0.305	0.299	0.440
20	0.333	0.381	0.040	0.331	0.372	**0.460**
25	0.330	0.283	0.237	0.345	0.409	**0.448**
30	0.333	0.413	0.228	0.368	0.405	**0.446**
35	0.297	0.409	0.199	0.364	0.348	**0.425**
40	0.340	**0.421**	0.288	0.381	0.376	0.408
45	0.393	0.386	0.333	0.375	**0.418**	0.360
50	0.422	**0.432**	0.328	0.375	0.396	0.341

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.231	-0.003	-0.064	0.191	0.205	**0.240**
10	0.364	-0.126	-0.068	0.254	0.305	**0.383**
15	**0.456**	0.126	-0.009	0.301	0.296	0.440
20	0.333	0.380	0.040	0.328	0.370	**0.460**
25	0.329	0.282	0.231	0.343	0.407	**0.448**
30	0.333	0.412	0.226	0.367	0.404	**0.446**
35	0.297	0.409	0.197	0.363	0.347	**0.425**
40	0.340	**0.420**	0.286	0.380	0.376	0.408
45	0.393	0.386	0.329	0.374	**0.418**	0.360
50	0.422	**0.432**	0.324	0.375	0.396	0.341

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: Weighted nonzero sampling, up to 400/student

Retained exfr edges: 290,800 / 1,167,562 (24.9%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.222	-0.010	-0.064	0.178	0.187	**0.238**
10	0.359	-0.133	-0.068	0.236	0.289	**0.379**
15	**0.450**	0.115	-0.030	0.281	0.281	0.438
20	0.328	0.369	0.020	0.307	0.352	**0.458**
25	0.324	0.271	0.172	0.322	0.390	**0.446**
30	0.327	0.403	0.166	0.343	0.387	**0.444**
35	0.292	0.402	0.129	0.340	0.330	**0.423**
40	0.336	**0.415**	0.214	0.360	0.363	0.407
45	0.392	0.381	0.248	0.351	**0.408**	0.359
50	0.421	**0.430**	0.262	0.354	0.389	0.340

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.209	-0.025	-0.064	0.154	0.151	**0.225**
10	0.337	-0.151	-0.074	0.204	0.243	**0.356**
15	**0.418**	0.090	-0.067	0.246	0.227	0.408
20	0.293	0.339	-0.028	0.262	0.290	**0.425**
25	0.284	0.234	0.107	0.274	0.326	**0.410**
30	0.282	0.354	0.090	0.296	0.317	**0.407**
35	0.245	0.356	0.055	0.290	0.254	**0.385**
40	0.288	0.358	0.133	0.312	0.278	**0.366**
45	**0.343**	0.336	0.163	0.306	0.321	0.315
50	0.376	**0.381**	0.169	0.311	0.304	0.295

Group 3: Weighted nonzero sampling, up to 200/student

Retained exfr edges: 145,400 / 1,167,562 (12.5%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.860	0.824	**0.900**	0.853	0.866	0.858
10	0.859	0.824	**0.877**	0.853	0.866	0.864
15	0.859	0.823	**0.877**	0.854	0.865	0.863
20	0.861	0.813	**0.876**	0.854	0.864	0.862
25	0.863	0.819	**0.875**	0.853	0.864	0.864
30	0.862	0.821	**0.875**	0.852	0.864	0.861
35	0.863	0.823	**0.876**	0.852	0.864	0.860
40	0.864	0.827	**0.875**	0.852	0.865	0.859
45	0.864	0.832	**0.874**	0.852	0.863	0.859
50	0.864	0.828	**0.874**	0.852	0.862	0.859

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	1.000	**1.000**	1.000	**1.000**	1.000	**1.000**
10	1.000	**1.000**	0.999	1.000	1.000	**1.000**
15	1.000	**1.000**	0.998	1.000	1.000	**1.000**
20	1.000	1.000	0.998	1.000	1.000	**1.000**
25	1.000	1.000	0.998	1.000	1.000	**1.000**
30	1.000	**1.000**	0.999	1.000	1.000	1.000
35	1.000	**1.000**	0.999	1.000	1.000	1.000
40	1.000	**1.000**	0.999	1.000	1.000	1.000
45	1.000	**1.000**	0.999	1.000	1.000	1.000
50	1.000	1.000	0.993	1.000	1.000	**1.000**

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: Weighted nonzero sampling, up to 200/student

Retained exfr edges: 145,400 / 1,167,562 (12.5%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.308**	0.031	-0.064	0.189	0.234	0.254
10	**0.388**	-0.079	-0.087	0.254	0.289	0.343
15	0.341	0.056	-0.125	0.289	0.338	**0.413**
20	0.333	0.304	0.072	0.321	**0.340**	0.289
25	0.360	0.309	0.017	0.342	0.331	**0.461**
30	0.378	0.412	0.120	0.343	0.318	**0.529**
35	0.386	**0.494**	0.226	0.362	0.315	0.391
40	0.378	**0.452**	0.278	0.370	0.301	0.329
45	0.342	0.368	**0.399**	0.375	0.281	0.290
50	0.348	**0.430**	0.356	0.374	0.349	0.386

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.306**	0.029	-0.064	0.184	0.231	0.249
10	**0.386**	-0.080	-0.087	0.249	0.285	0.339
15	0.340	0.055	-0.132	0.285	0.336	**0.409**
20	0.332	0.303	0.065	0.318	**0.337**	0.286
25	0.359	0.308	0.014	0.340	0.329	**0.459**
30	0.378	0.411	0.087	0.341	0.317	**0.528**
35	0.385	**0.494**	0.196	0.360	0.312	0.390
40	0.378	**0.452**	0.234	0.369	0.299	0.329
45	0.342	0.368	0.357	**0.374**	0.279	0.290
50	0.348	**0.430**	0.326	0.374	0.348	0.385

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: Weighted nonzero sampling, up to 200/student

Retained exfr edges: 145,400 / 1,167,562 (12.5%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.297**	0.021	-0.064	0.172	0.219	0.236
10	**0.375**	-0.089	-0.087	0.237	0.273	0.323
15	0.324	0.043	-0.150	0.272	0.323	**0.390**
20	0.319	0.287	0.028	0.301	**0.324**	0.275
25	0.348	0.294	-0.003	0.325	0.319	**0.450**
30	0.367	0.396	0.028	0.326	0.296	**0.521**
35	0.375	**0.483**	0.119	0.346	0.298	0.384
40	0.369	**0.445**	0.153	0.354	0.288	0.324
45	0.334	0.361	0.261	**0.362**	0.269	0.285
50	0.341	**0.425**	0.243	0.362	0.339	0.381

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.264**	0.007	-0.064	0.157	0.186	0.201
10	**0.336**	-0.107	-0.093	0.210	0.220	0.282
15	0.275	0.019	-0.179	0.234	0.264	**0.337**
20	0.261	0.253	-0.036	0.258	**0.263**	0.235
25	0.289	0.252	-0.067	0.277	0.243	**0.413**
30	0.305	0.344	-0.047	0.270	0.202	**0.485**
35	0.306	**0.426**	0.038	0.290	0.211	0.347
40	0.299	**0.393**	0.067	0.297	0.203	0.286
45	0.259	0.296	0.161	**0.302**	0.181	0.247
50	0.267	**0.362**	0.153	0.301	0.258	0.344